

End-of-outbreak decision-making under superspreading and onset-date delays

The 2018 Équateur Province (DRC) Ebola outbreak

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Abstract

Deciding when an outbreak is over requires estimating the probability that further cases will occur if control measures are relaxed. This estimate depends on two often-implicit choices: how transmission heterogeneity is represented and whether symptom-onset dates are treated as if they were infection dates. We compared five renewal models of the 2018 Équateur Province Ebola outbreak, all calibrated to the same onset-to-onset serial interval, and calculated the risk of additional cases (RAC) after each day. Models placing overdispersion at the level of the day, transmission event, or individual produced materially different risks at the same nominal dispersion parameter and inferred different values when that parameter was estimated. Models that anchored transmission to the infector’s symptom onset and represented incubation explicitly crossed both decision thresholds 8–12 days earlier than their infection-anchored counterparts; they also inferred substantially smaller post-intervention reproduction numbers. Finally, distinguishing additional cases from additional transmission isolated the contribution of people already infected but not yet symptomatic—a component that infection-anchored models cannot represent. End-of-outbreak assessments therefore depend not only on the amount of transmission heterogeneity, but also on its mechanism and on the time scale to which transmission is anchored.

1 Introduction

An end-of-outbreak declaration is a decision under uncertainty: interventions are withdrawn when the probability of further cases is judged acceptably small. Renewal models provide a natural basis for that decision by linking observed incidence to past transmission and projecting forward under relaxed control. Two choices within these models are seldom examined together.

The mechanism of overdispersion. Transmission of Ebola virus disease, like that of many pathogens, is highly heterogeneous, and this is usually represented by a single dispersion parameter k . Models can share a mean–variance relationship at the population level while differing in *where* the excess variance sits — in a day’s aggregate incidence, in each transmission event, or in each infected individual. Those are different statements about the world, and we show they imply different end-of-outbreak risks. They also make k a different quantity in each case, so a literature estimate obtained under one mechanism is not portable to another. In particular, an individual-level estimate such as $k = 0.18$ does not retain its interpretation in a day-level incidence model.

Onset dates are not infection dates. Renewal models are routinely fitted with a symptom-onset time series substituted directly for an infection time series. This conflates two distinct events separated by an incubation period, which for Ebola virus disease has a mean of about 11.4 days. We quantify the consequences by comparing infection-anchored models with explicitly onset-anchored counterparts, in which transmission is referenced to the infector’s own onset and the incubation period is modelled.

We compare five models using the 2018 Équateur Province Ebola outbreak. All use the same onset-to-onset serial interval, allowing differences to be attributed to transmission mechanism and temporal anchoring rather than to different interval assumptions. We first compare three infection-anchored mechanisms with the dispersion fixed and estimated, then compare event- and individual-level models with onset-anchored counterparts. The primary estimand is the risk of additional cases; a supplementary analysis separates it from the risk of additional transmission. Sensitivity analyses needed to assess the robustness of these comparisons are specified separately.

2 Data and delay distributions

2.1 The 2018 Équateur outbreak

The observed series is a daily count of Ebola virus disease cases by *symptom-onset date* for the 2018 Équateur Province outbreak in the Democratic Republic of the Congo: 54 cases, with onsets on 31 distinct days between 5 April 2018 and 2 June 2018, and at most 6 onsets on any one day. An Ebola Response Team (ERT) was present from 8 May 2018 and withdrew on 24 July 2018. Indexing days from the first observed onset:

	Day index	Date
First onset (index case)	0	5 April 2018
ERT arrival	33	8 May 2018
Last observed onset	58	2 June 2018
ERT withdrawal (window end)	110	24 July 2018

The analysis window is days 0–110, i.e. 111 days, with 28 cases before the ERT arrived and 26 after. The series is padded with zero counts from the last observed onset to the end of the window. These case-free days provide the evidence that drives the estimated risk down. No adjustment is made for under-reporting.

Day 0 is treated as an initial condition in every model, so likelihoods run over days $1, \dots, 110$. Conditioning on a single imported index case asserts that nothing was transmitting or incubating beforehand. For the infection-anchored models this is mild — the renewal recursion simply starts from one case. For the onset-anchored models it is stronger, because it also asserts an *empty incubation pipeline* at day 0: anyone infected shortly before the index case’s onset would contribute onsets in the first weeks that the model can explain only as transmission from the index case, inflating R_{pre} . The onset-anchored models are therefore more sensitive to whether the first observed case is the true epidemiological index case.

2.2 Serial interval

We use a generic Ebola serial-interval estimate, represented by a discretised gamma distribution with mean 15.3 d and standard deviation 9.3 d, in all five models. An outbreak-specific estimate fitted to contact-traced pairs from this outbreak has mean 19.46 d and standard deviation 6.08 d.

We selected the generic estimate because it is compatible with the variance budget of Section 2.3. Splitting an onset-to-onset interval into an incubation period and a time-from-onset-to-transmission (TOST) delay requires the two variances to sum to the serial interval’s. The generic estimate has variance 86.49 d², which comfortably accommodates standard published Ebola incubation estimates. The outbreak-specific estimate has variance 36.97 d², *smaller* than the incubation variance alone (65.61 d²), and is therefore structurally inadmissible for the onset-anchored models under the assumed independent delay decomposition. Sensitivity to the outbreak-specific estimate can therefore be assessed only for the infection-anchored models.

2.3 Discretisation, indexing and the decomposition

Continuous delays are gamma distributed and discretised by the method of Cori et al. (2013): the mass at integer lag s is $\int_{s-1}^{s+1} (1 - |s - u|) g(u) du$, normalised. Three discrete delays are used, and they differ in the lag at which they start:

Distribution	Weights	First lag	Interpretation
Serial interval	w_s	1	onset of infector $\rightarrow$ onset of infectee
Incubation period	f_a^{inc}	1	infection $\rightarrow$ onset
Onset to transmission	f_s^{tost}	0	onset of infector $\rightarrow$ transmission

Write $F_r = \sum_{s=1}^r w_s$ with $F_0 = 0$ for the serial-interval distribution function, and let

$$\bar{f}_r^{\text{tost}} = \Pr(S > r) = 1 - \sum_{s=0}^r f_s^{\text{tost}}, \quad \bar{f}_r^{\text{inc}} = \Pr(A > r) = 1 - \sum_{a=1}^r f_a^{\text{inc}} \quad (1)$$

be the survival functions of the TOST delay S and the incubation period A . Note $\bar{f}_0^{\text{inc}} = 1$ but $\bar{f}_0^{\text{tost}} = 1 - f_0^{\text{tost}} \leq 1$: a case has all of its incubation ahead of it on the day it is infected, but has already had its lag-0 transmission opportunity on the day it becomes symptomatic.

The lag-0 rule is what makes the onset-anchored recursion causal. Because f^{inc} places no mass at lag 0, an infection on day t cannot produce an onset before day $t + 1$, so a day's onset cohort is fully determined before that day's own transmission is evaluated. The TOST distribution *may* place mass at lag 0, so a case can transmit on its own onset day. At most one of the two may be supported at lag 0: if both were, infection, onset and onward transmission could all fall within one day and the recursion would not be well ordered.

The variance budget. For an infector with onset on day u , transmission occurs on day $u + S$ and the infectee's onset on day $u + S + A$, so the onset-to-onset interval is $S + A$ and matching the target serial interval requires

$$\mathbb{E}[S] + \mathbb{E}[A] = 15.3, \quad \text{Var}[S] + \text{Var}[A] = 86.49. \quad (2)$$

The incubation period is fixed from a published Ebola estimate — mean 11.4 d, standard deviation 8.1 d (WHO Ebola Response Team, 2014) — and f^{tost} is the gamma carrying the residual mean and variance, here mean 3.9 d and standard deviation 4.57 d. A negative residual variance would make the incubation estimate inadmissible against the serial interval. Discretising and then convolving is not quite the same operation as convolving and then discretising, so we verified the decomposition numerically: the discrete convolution $f^{\text{tost}} * f^{\text{inc}}$ differs from the discretised target w by 0.00070 in total variation. All delays are truncated at lag 110, the length of the analysis window, so nothing is truncated within it.

Remark. The convolution is faithful, but *how* the mean and variance are split between incubation and TOST is assumed rather than identified by the serial interval alone. The sensitivity of the onset-anchoring results to this decomposition must therefore be assessed explicitly.

3 Models

3.1 Notation

Let C_t be the observed count on day t . The three *infection-anchored* models set $I_t := C_t$, reading onset dates as infection dates — the practice under study. The two *onset-anchored* models set $D_t := C_t$ and treat onsets as onsets, with infections and onsets separated by the incubation period. The reproduction number R_t takes the value R_{pre} before the ERT arrives and R_{post} from its arrival onwards (Section 3.5), and $k > 0$ is a dispersion parameter, small k corresponding to high heterogeneity.

Remark. $\text{NB}(\text{mean} = \mu, \text{disp} = \kappa)$ denotes the negative binomial with mean μ , variance $\mu + \mu^2/\kappa$ and probability generating function $G(s) = (\kappa/(\kappa + \mu(1-s)))^\kappa$; in particular $G(0) = (1 + \mu/\kappa)^{-\kappa}$. $\text{Gamma}(\alpha, \beta)$ has shape α , rate β and mean α/β .

3.2 Infection-anchored models

The Poisson baseline and the three compared infection-anchored models share one generative skeleton. Cases beget cases through the serial interval: an individual infected on day u produces secondary infections that appear on day $u + \ell$ with probability w_ℓ , and the reproduction number in force on the day a secondary case appears sets their expected number. Write

$$\lambda_t = \sum_{s \geq 1} w_s I_{t-s} \quad (3)$$

for the force of infection driving day t . What distinguishes the models is the offspring law — where, if anywhere, the excess variance sits.

From transmission to observations. Infection and symptom onset are identified, so $I_t = C_t$. The transmission law for I_t is therefore also the likelihood for the observed daily counts; no incubation step intervenes.

Poisson baseline: Cori. An individual infected on day u generates, at each lag ℓ and independently of every other, a Poisson number of offspring with mean $R_{u+\ell}w_\ell$. Poisson superposition then gives the day- t total as

$$\text{Cori :} \quad I_t \mid \lambda_t \sim \text{Poisson}(R_t \lambda_t). \quad (4)$$

This is the standard renewal model of Cori et al. (2013) and the model underlying most published end-of-outbreak risk calculations. It has no dispersion parameter.

Transmission-event heterogeneity: SSE. The offspring law is negative binomial, with a dispersion that *does* scale with the force of infection:

$$\text{SSE :} \quad I_t \mid \lambda_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t). \quad (5)$$

That scaling is what makes k an *individual-level* offspring dispersion. Each infector independently contributes $\text{NB}(R_t w_\ell, k w_\ell)$ offspring at lag ℓ ; these all share the success probability $k/(k + R_t)$, which does not depend on ℓ , so by negative binomial closure they pool into exactly (5), and each individual's own offspring total is $\text{NB}(R, k)$. The decomposition is used again in Section 4.2.

Equivalently, (5) has the Poisson–Gamma representation

$$\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k \lambda_t, k), \quad I_t \mid \tilde{\lambda}_t \sim \text{Poisson}(R_t \tilde{\lambda}_t), \quad (6)$$

in which the day's transmission opportunity is inflated or deflated by a fresh Gamma factor. The marginal form (5) is used directly for SSE; the mixture representation is needed to define its onset-anchored counterpart below.

Individual heterogeneity: SSI. Heterogeneity is attached to people rather than to occasions. Each infected individual carries a latent infectivity drawn independently from $\text{Gamma}(k, k)$, with mean 1 and variance $1/k$, so the aggregate infectivity of the day- t cohort is $\text{Gamma}(k I_t, k)$ and infections arise as a Poisson process driven by past infectivity:

$$\text{SSI :} \quad Y_t \mid I_t \sim \text{Gamma}(k I_t, k), \quad I_t \mid (Y_u)_{u < t} \sim \text{Poisson}(R_t \lambda_t^Y), \quad \lambda_t^Y = \sum_{s \geq 1} w_s Y_{t-s}. \quad (7)$$

Here $I_t = 0$ forces $Y_t = 0$. Marginally each individual's offspring total is $\text{NB}(R, k)$, the same as under SSE — the two models are calibrated to the same individual-level offspring distribution and differ only in mechanism. But the Y_t cannot be integrated out of the joint law and must be inferred with the other model parameters.

Day-level heterogeneity: DLO. Unlike the three families above, DLO has no onset-anchored counterpart in this study. The negative binomial is applied directly to the day's aggregate incidence, with the same dispersion on every day:

$$\text{DLO : } I_t \mid \lambda_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k). \quad (8)$$

Its excess variance is a property of the day and does not scale with the number of infectious people. Thus k describes day-to-day variation in aggregate incidence rather than inequality in individual transmission, and the same numerical value has a different meaning here than in SSE or SSI.

Remark (The Poisson limit). As $k \rightarrow \infty$ the negative binomial in (5) converges to the Poisson and the Gamma noise in (7) degenerates to $Y_t = I_t$, so SSE and SSI both collapse onto Cori. DLO loses its excess variance in the same limit, but without the parameter ever having had an individual-level reading to lose.

3.3 Onset-anchored models

Transmission is now referenced to the infector's own symptom onset rather than to their unobserved infection time, which is appropriate for a disease with negligible pre-symptomatic transmission. Two things change. The serial interval is replaced by the TOST distribution, so the force of infection is a weighted sum of past *onset* cohorts; and infection and onset become distinct events, separated by the incubation period. Each of the three models below is the exact onset-anchored counterpart of one of (4), (5) and (7), and is presented in the same order. DLO is omitted because no onset-anchored day-level model is compared.

From transmission to observations. Let J_t denote the number of infections on day t and $D_t = C_t$ the number of observed onsets. In all three models, day- u infections are carried forward to their onset days by independent incubation draws,

$$(N_{u,a})_{a \geq 1} \mid J_u \sim \text{Multinomial}(J_u, (f_a^{\text{inc}})), \quad D_t = \sum_{a \geq 1} N_{t-a,a}, \quad (9)$$

and only the onsets are observed. The infection counts J_t and allocations $N_{u,a}$ can be integrated out exactly.

Proposition 1 (Onsets alone suffice). *Suppose that, conditional on a non-negative day- t rate E_t that is a function of $D_{0:t}$ and of any latent variables attached to days $u \leq t$, the infections satisfy $J_t \mid E_t \sim \text{Poisson}(E_t)$, and that they are allocated to onset days by (9). Then that model and*

$$D_t \mid (E_u)_{u < t} \sim \text{Poisson}\left(\sum_{a \geq 1} f_a^{\text{inc}} E_{t-a}\right) \quad (10)$$

define the same joint law for the observed onset process (D_t) .

The proof is in Appendix A. It rests on Poisson thinning, which turns the multinomial split (9) into independent Poisson parts, and on Poisson superposition, which recombines them into (10); the same thinning argument supplies the incubation pipeline of Section 4.3. The hypothesis covers all three models: two have Poisson offspring directly, and SSE-SO's negative binomial is Poisson conditional on its Gamma factor, as in (6). We therefore give both the infection process and the equivalent likelihood for observed onsets.

Poisson baseline: Cori-SO. A case with onset on day u infects others s days later with probability f_s^{tost} , in $\text{Poisson}(R_{u+s}f_s^{\text{tost}})$ numbers and independently across lags and individuals. Summing over infectors,

$$\text{Cori-SO :} \quad J_t \mid D_{0:t} \sim \text{Poisson}(R_t \lambda_t), \quad \lambda_t = \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}, \quad (11)$$

together with (9). Proposition 1 applies with $E_t = R_t \lambda_t$ and reduces this to

$$D_t \mid D_{0:t-1} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \sum_{s \geq 0} f_s^{\text{tost}} D_{t-a-s} \right), \quad (12)$$

a deterministic recursion in the observed onsets.

Transmission-event heterogeneity: SSE-SO. The offspring law is negative binomial with a dispersion scaling with the force of infection, as in (5):

$$\text{SSE-SO :} \quad J_t \mid D_{0:t} \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k \lambda_t), \quad \lambda_t = \sum_{s \geq 0} f_s^{\text{tost}} D_{t-s}, \quad (13)$$

together with (9). Writing the negative binomial as a Poisson–Gamma mixture puts a fresh Gamma factor on each day’s transmission opportunity, and Proposition 1 then applies with $E_t = R_t \tilde{\lambda}_t$:

$$\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k \lambda_t, k), \quad D_t \mid (\tilde{\lambda}_u)_{u < t} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \tilde{\lambda}_{t-a} \right). \quad (14)$$

Here, unlike in SSE, the Gamma factor cannot be integrated out again: each D_t pools contributions from many days, each carrying its own independent $\tilde{\lambda}$, so the onsets are *not* marginally negative binomial. The day-specific Gamma effects must therefore be retained in the likelihood.

Individual heterogeneity: SSI-SO. Each case carries a latent infectivity drawn from $\text{Gamma}(k, k)$, so the day- t onset cohort has aggregate infectivity $\text{Gamma}(k D_t, k)$ and drives a Poisson infection process, as in (7):

$$\begin{aligned} \text{SSI-SO :} \quad Y_t \mid D_t &\sim \text{Gamma}(k D_t, k), \quad J_t \mid (Y_u)_{u \leq t} \sim \text{Poisson}(R_t \lambda_t^Y), \\ \lambda_t^Y &= \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-s}, \end{aligned} \quad (15)$$

together with (9). Proposition 1 applies with $E_t = R_t \lambda_t^Y$ and gives

$$Y_t \mid D_t \sim \text{Gamma}(k D_t, k), \quad D_t \mid (Y_u)_{u < t} \sim \text{Poisson} \left(\sum_{a \geq 1} f_a^{\text{inc}} R_{t-a} \sum_{s \geq 0} f_s^{\text{tost}} Y_{t-a-s} \right). \quad (16)$$

The latents are the same cohort infectivities as under SSI and, as there, cannot be integrated out.

Remark (The Poisson limit). As in Remark 3.2, $k \rightarrow \infty$ collapses SSE-SO and SSI-SO onto Cori-SO. Neither Cori nor Cori-SO is one of the five compared models. They provide the Poisson limits against which the overdispersed models are interpreted; Cori also links the analysis to a published Équateur risk curve.

3.4 The five compared models

Model	Anchoring	Excess variance acts at the level of	Latents
DLO	infections [†]	the day — NB on aggregate incidence	none
SSE	infections [†]	the transmission event	none
SSI	infections [†]	the individual	Y_t
SSE-SO	symptom onsets	the transmission event	$\tilde{\lambda}_t$
SSI-SO	symptom onsets	the individual	Y_t

[†]Observed onsets are treated as infection dates.

3.5 When does R switch?

R switches from R_{pre} to R_{post} on day 33 (8 May 2018) **in each model’s own time index**. In the onset-anchored models this is unambiguous: R_t multiplies the TOST-weighted force of infection producing *infections* on day t , so the intervention acts on transmission from that date and the resulting onsets appear an incubation period later. In the infection-anchored models R_t multiplies the force of infection producing *cases appearing* on day t , so the switch attributes the intervention’s effect to onsets from that date.

The displacement between the two conventions is the infectee’s **incubation period**, not the serial interval. A case with onset on day t was infected on day $t - A$ with $A \sim f^{\text{inc}}$; the infection-anchored model dates the intervention’s effect on that transmission to t , the onset-anchored model to $t - A$. The relevant delay is therefore $\mathbb{E}[A] = 11.4$ d. (The serial interval, 15.3 d, is the infector-onset to infectee-onset gap and does not separate the two conventions.)

This asymmetry is a consequence of onset-as-infection conflation: it is one route by which infection-anchored models mis-date an intervention’s effect. Section 6.3 quantifies its consequences for R_{post} and the risk curve.

4 The risk of additional cases and of additional transmission

4.1 Definition

Writing the estimand as $\Pr(\cdot \mid \text{data up to day } t)$ invites a filtering reading that the analysis does not use, so we state it as a procedure.

Fit $(R_{\text{pre}}, R_{\text{post}}, k)$ and all latent states to the complete record, days 0–110. Then, for each day t : retain the inferred state attached to the observed history through day t , discard the realised trajectory after day t , reset R to $R^* = R_{\text{pre}}$ from day $t + 1$ onwards, and simulate a counterfactual future. $\text{RAC}(t)$ is the posterior probability that this replicated future contains at least one further case.

Two things follow that the shorthand hides. The fit is fully retrospective by design: the parameters come from all 111 days, not from the history through t . And for the latent-variable models the *retained state* also comes from the full-data posterior, so it is a *smoothed* state informed by data after day t (Section 4.5).

Under the onset-anchored models a second quantity becomes distinct:

- $\text{RAC}(t)$, the probability of at least one further *case*, i.e. a further entry in the onset series after day t . It counts individuals who are already infected on day t and become symptomatic afterwards, even if no new transmission occurs, so it does not approach zero until the incubation pipeline has drained. This is the directly observable quantity and the headline throughout.
- $\text{RAT}(t)$, the probability of at least one further *transmission event* after day t . It isolates the effect of relaxing interventions but is not directly observable.

Under the three infection-anchored models an infection and the case it produces fall on the same day by construction, so the two coincide *by assumption* — which is precisely the conflation under study. RAT is therefore reported only in the supplementary comparison of Section 6.4.

RAC(t) is one number per day, not a distribution: it is a posterior probability, so the average over draws happens inside it. In each case below we obtain a conditional zero-probability given one posterior draw, and report

$$\text{RAC}(t) = 1 - \frac{1}{S} \sum_{m=1}^S \exp(\log \Pr(\text{no further case} \mid \theta^{(m)})), \quad (17)$$

over S posterior draws, and likewise for RAT. Because this is an average it carries Monte-Carlo error, which is estimated from the between-chain spread and reported beside every curve.

4.2 Infection-anchored models

Fix one posterior draw. In the infection-anchored models, the observed incidence and, for SSI, the retained infectivities completely specify the reset state: there is no separate incubation pipeline.

Retained transmission weight. Define

$$\Lambda(t) = \sum_{u \leq t} I_u (1 - F_{t-u}) \quad (\text{Cori, SSE, DLO}), \quad \Lambda_Y(t) = \sum_{u \leq t} Y_u (1 - F_{t-u}) \quad (\text{SSI}), \quad (18)$$

where $(1 - F_{t-u})$ is the probability that a case first appearing on day u still has transmission to come after day t ; with $F_0 = 0$, the $u = t$ term enters with full weight.

Future-event exponent. Let $H_{\text{inf}}(t)$ be minus the log-probability that the retained state produces no future event:

$$H_{\text{inf}}(t) = \begin{cases} R^* \Lambda(t) & (\text{Cori}), \\ k \Lambda(t) \log(1 + R^*/k) & (\text{SSE}), \\ R^* \Lambda_Y(t) & (\text{SSI}), \\ k \sum_{j>t} \log(1 + R^* \mu_j/k) & (\text{DLO}), \end{cases} \quad \mu_j = \sum_{s \geq 1} w_s I_{j-s}, \quad (19)$$

where the future counts in μ_j are set to zero, $I_u := 0$ for $u > t$.

Proposition 2 (Conditional zero-probabilities, infection-anchored models). *Conditional on one draw of the parameters and, for SSI, the retained latent infectivities,*

$$\log \Pr(\text{no further case after } t) = -H_{\text{inf}}(t), \quad (20)$$

$$\log \Pr(\text{no further transmission after } t) = -H_{\text{inf}}(t). \quad (21)$$

Proof. Retained cohorts. For Cori, each observed case contributes $\text{Poisson}(R^* w_\ell)$ offspring at lag ℓ independently; the contributions with $\ell > t - u$ are untouched by the conditioning and pool by Poisson superposition into $\text{Poisson}(R^* \Lambda(t))$. For SSE, the same contributions are $\text{NB}(R^* w_\ell, k w_\ell)$ and pool by negative binomial closure into $\text{NB}(R^* \Lambda(t), k \Lambda(t))$, whose zero-probability is as stated. For SSI, given the latents they are again Poisson, with rates $R^* w_\ell Y_u$, pooling into $\text{Poisson}(R^* \Lambda_Y(t))$. For DLO, the future counts are negative binomial day by day with a dispersion that does not depend on the force of infection, so the zero-probabilities multiply over days. Taking minus the logarithm gives the four rows of (19).

Cases versus transmission. Because infection and symptom onset are identified in these models, a future transmission event and the future case it creates are represented as occurring together. $\square$

Thus $\text{RAC}(t) = \text{RAT}(t) = 1 - e^{-H_{\text{inf}}(t)}$. The Cori row is included as the Poisson reference, not as a compared model. It is the $k \rightarrow \infty$ limit of the SSE and SSI rows, with $\Lambda_Y(t) \rightarrow \Lambda(t)$. The DLO sum is finite because $\mu_j = 0$ once j exceeds t plus the support of w .

4.3 Onset-anchored models

Fix one posterior draw. Here the observed onsets do not completely specify the reset state: in addition to retained onset cohorts, it contains a latent pipeline of individuals already infected but not yet symptomatic. Write R_u for the historical reproduction number on day u .

Retained transmission weight. The driving series still able to transmit after t ,

$$W(t) = \sum_{u \leq t} \mathcal{D}_u \bar{f}_{t-u}^{\text{tost}}, \quad \mathcal{D}_u = \begin{cases} D_u & (\text{Cori-SO}), \\ D_u & (\text{SSE-SO}), \\ Y_u & (\text{SSI-SO}), \end{cases} \quad (22)$$

the onset-anchored analogue of $\Lambda(t)$, with the TOST survival function of (1) in place of $1 - F$.

Incubation pipeline mean. The expected number infected on or before day t whose onset falls after it,

$$M(t) = \sum_{u \leq t} E_u \bar{f}_{t-u}^{\text{inc}}, \quad (23)$$

with E_u the model's conditional mean number of infections on day u — $R_u \lambda_u$ for Cori-SO, $R_u \tilde{\lambda}_u$ for SSE-SO and $R_u \lambda_u^Y$ for SSI-SO, as in (12)–(16) — evaluated at the *historical* R_u : the pipeline was filled before the reset.

Future-event exponent. Let c be minus the log-probability that one future case, transmitting under R^* over its whole TOST support, has no offspring:

$$c = \begin{cases} R^* & (\text{Cori-SO}), \\ k \log(1 + R^*/k) & (\text{SSE-SO}), \\ k \log(1 + R^*/k) & (\text{SSI-SO}). \end{cases} \quad (24)$$

The exponent for no transmission from onset cohorts already observed by day t is

$$H_{\text{on}}(t) = \begin{cases} R^* W(t) & (\text{Cori-SO}), \\ c W(t) & (\text{SSE-SO}), \\ R^* W(t) & (\text{SSI-SO}). \end{cases} \quad (25)$$

For SSI-SO, $H_{\text{on}}(t)$ conditions on the inferred infectivities of onset cohorts already observed, whereas c averages over the as-yet-unobserved infectivity of a future pipeline member.

Proposition 3 (Conditional zero-probabilities, onset-anchored models). *Conditional on one draw of the parameters and the retained latent state,*

$$\log \Pr(\text{no further case after } t) = -H_{\text{on}}(t) - M(t), \quad (26)$$

$$\log \Pr(\text{no further transmission after } t) = -H_{\text{on}}(t) - M(t)(1 - e^{-c}). \quad (27)$$

Proof. Two independent sources of future events must be accounted for: the onset cohorts retained through day t , which have transmission opportunity left, and the pipeline of individuals already infected but not yet symptomatic.

Retained cohorts. Under Cori-SO and SSI-SO the infections generated by cohort u at TOST lag s are independent $\text{Poisson}(R^* f_s^{\text{tost}} \mathcal{D}_u)$; those with $s > t - u$ are untouched by the conditioning, and summing over s and $u \leq t$ gives $\text{Poisson}(R^* W(t))$, with zero-probability $e^{-R^* W(t)}$. Under SSE-SO the same contributions are $\text{NB}(R^* f_s^{\text{tost}} \mathcal{D}_u, k f_s^{\text{tost}} \mathcal{D}_u)$ with common success probability $k/(k + R^*)$, so they pool into $\text{NB}(R^* W(t), kW(t))$, whose zero-probability is $(1 + R^*/k)^{-kW(t)} = e^{-cW(t)}$. Either way the probability is $e^{-H_{\text{on}}(t)}$.

Pipeline. Poisson splitting of the day- u infections by the incubation delay (the argument of Appendix A) makes the number infected on or before t but symptomatic after t Poisson with mean $M(t)$, independently of the retained cohorts' remaining transmission.

Cases versus transmission. A future *case* requires either a pipeline member to become symptomatic or a retained cohort to generate a new infection — and every new infection produces an onset strictly later, since f^{inc} has no mass at lag 0. So no further case has probability $e^{-M(t)} e^{-H_{\text{on}}(t)}$, giving (26).

A future *transmission event* permits pipeline members to become symptomatic provided each transmits zero times, which each does independently with probability e^{-c} by (24). If the pipeline size is $N \sim \text{Poisson}(M(t))$ then $\mathbb{E}[(e^{-c})^N] = \exp(-M(t)(1 - e^{-c}))$ by the Poisson generating function, and multiplying by $e^{-H_{\text{on}}(t)}$ gives (27). $\square$

Comparison of the two anchorings. Both $H_{\text{inf}}(t)$ and $H_{\text{on}}(t)$ describe transmission still owed by cohorts observed through day t . The onset-anchored calculation adds $M(t)$, the infections already generated under the historical R_u whose onsets lie after t . A further *case* requires that pipeline to be empty; no further *transmission* requires only that its members have no offspring. Setting $M(t) = 0$ reduces the onset-anchored expressions to the same retained-cohort structure as their infection-anchored counterparts, with the TOST survival replacing the serial-interval survival.

Remark (Immediate consequences). Three special cases follow immediately and are used as verification targets (Section 5.6).

- (i) Since $1 - e^{-c} \leq 1$, (27) is never below (26), so $\text{RAC}(t) \geq \text{RAT}(t)$ always, with equality exactly when $M(t) = 0$. The gap is the pipeline, and it is the term the infection-anchored models set to zero by assumption.
- (ii) Setting $R^* = 0$ gives $c = 0$ and $H_{\text{on}}(t) = 0$, so $\text{RAT}(t) = 0$ exactly and $\text{RAC}(t) = 1 - e^{-M(t)}$: with no future transmission permitted, the only remaining risk is that the retained pipeline is non-empty.
- (iii) As $k \rightarrow \infty$, $c = k \log(1 + R^*/k) \rightarrow R^*$ and SSE-SO's $H_{\text{on}}(t)$ tends to $R^*W(t)$, so both overdispersed models collapse onto Cori-SO.

Remark (Why historical and reset reproduction numbers both appear). $H_{\text{on}}(t)$ uses the reset R^* , because it describes transmission that has yet to happen; $M(t)$ uses the historical R_u , because the pipeline it counts was generated before the reset. Under the infection-anchored models there is no pipeline, so the whole retained state is projected at R^* . This is the formal counterpart of the timing argument of Section 6.3.

4.4 What the three mechanisms do differently

Proposition 2 makes the structural comparison explicit. SSE and SSI depend on the observed history *only through the scalar* $\Lambda(t)$ or $\Lambda_Y(t)$: the total remaining offspring pools into a single negative binomial or, given the latents, a single Poisson. DLO does not. Its risk depends on the whole temporal *profile* $(\mu_j)_{j>t}$ of the future force of infection, because a fresh dispersion that does not scale with that force is applied on each future day. Writing $\varphi(\mu) = -k \log(1 + R^*\mu/k)$, which is convex with $\varphi(0) = 0$ and hence superadditive,

$$-R^*\Lambda(t) \leq \log \Pr(\text{no further case})_{\text{DLO}} = \sum_{j>t} \varphi(\mu_j) \leq \varphi(\Lambda(t)). \quad (28)$$

Spreading the same total $\Lambda(t)$ across days *dilutes* the overdispersion, so a value of k calibrated as an individual-level offspring dispersion does far less work in DLO than in SSE or SSI. The lower bound is the negative of the Cori exponent in (19), so DLO's risk is squeezed towards Cori's precisely when the future force of infection is thin and spread out.

Remark. The bounds (28) hold always; the *direction* of the comparison with SSE does not. It is a property of a thin, spread-out profile rather than a theorem, and concentrating $\Lambda(t)$ into a day or two reverses it.

4.5 Retrospective conditioning of latent states

The estimand is deliberately retrospective. For SSI, SSE-SO and SSI-SO, the state retained at day t is therefore drawn from its *smoothed* posterior conditional on the complete record, not from the filtering distribution conditional only on observations through t . Consequently, $\text{RAC}(t)$ should not be interpreted as a real-time probability for these models. Future observations can move the retained state in either direction: later cases imply greater infectivity in earlier cohorts, whereas later zeros imply less. We compare smoothed and filtering risks with particle methods as a sensitivity analysis. DLO and SSE have no latent state and are unaffected by this distinction.

4.6 Recovering integrated-out latent states

Some latent states can be integrated out of the likelihood analytically (Section 5.3) but are still needed to define the reset state. Before calculating either risk, we recover these states from their exact conditional distribution, $\text{Gamma}(k \text{ scale}_u, k + c_u)$, conditional on each posterior draw. This restores the complete latent trajectory, including infections after the final observed onset that may remain in the incubation pipeline.

5 Inference

5.1 Priors

We use proper priors, shared across models within each analysis, so that marginal likelihoods and posterior model probabilities are well defined.

- R_{pre} and R_{post} : independent log-normals with median 1.0 and 95% interval (0.2, 5.0).
- k , where estimated: an informative log-normal with median 0.18 and 95% interval (0.09, 0.36), identical in every model.
- Prior model probabilities: uniform over the models in each analysis, i.e. 1/3 in Analyses 1–2 and 1/4 in Analyses 3–4.

Remark (Identical priors are necessary but not sufficient for a fair comparison). Identical priors do not imply identical interpretations. In SSE and SSI, k is an individual-level offspring dispersion; in DLO, it controls day-level incidence variation (Section 4.4). Analysis 1 applies an individual-level literature estimate to all three models to test its portability. Its posterior model probabilities therefore compare the full model specifications, including that imposed value, and should not be read as evidence about mechanism alone.

5.2 Posterior sampling

All fits use Hamiltonian Monte Carlo (the No-U-Turn sampler, as implemented in PyMC), with 4 chains of 2000 draws after 2000 tuning iterations in every analysis, at a target acceptance probability of 0.90 in Analyses 1–3 and 0.95 in Analysis 4. One fit per model and analysis jointly estimates R_{pre} , R_{post} , any latent states, and — in Analyses 2 and 4 — k .

Convergence. Across all models of all four analyses, the worst $\hat{R}$ for the reported scalar parameters is 1.00, the smallest bulk effective sample size is 3575, and the total number of divergent transitions is 0. Per analysis the minimum bulk effective sample sizes are 4855, 4074, 4190 and 3575, with no divergences in any of them.

5.3 Latent-state sampling

The Gamma latent variables, particularly in SSE-SO, include very small shape parameters. We combine exact marginalisation with an inverse-CDF parameterisation to obtain stable posterior sampling without changing the model.

- **Exact marginalisation.** A latent that reaches no observation day carrying a case enters the likelihood only through an exponential factor linear in that latent. Its Gamma prior is conjugate to this factor, giving $\int \text{Gamma}(y; k \text{ scale}, k) e^{-cy} dy = (1 + c/k)^{-k \text{ scale}}$ and a conditional posterior $\text{Gamma}(k \text{ scale}, k + c)$. This removes 52 of SSE-SO’s 110 latents, leaving 58 sampled, and removes 1 of SSI’s and SSI-SO’s 31. Section 4.6 describes their exact recovery for the risk calculation.
- **Inverse-CDF reparameterisation.** The surviving latents are sampled as $\text{Uniform}(0, 1)$ variates transformed through the corresponding Gamma quantile function.

5.4 Model evidence

We compute the marginal likelihood $p(D_{1:110} \mid \text{model})$ by integrating over both parameters and latent states. Bridge sampling is the primary estimator; prior Monte Carlo and importance sampling provide independent numerical checks, and deterministic quadrature is used where the parameter space is sufficiently small. Posterior model probabilities follow from these marginal likelihoods and the uniform model prior of Section 5.1.

5.5 Comparing dispersion posteriors

Where k is estimated, we compare every pair of posterior distributions using three complementary quantities: the *median ratio*, which measures location; the *overlap* $\int \min(p_a, p_b)$, a scale-free symmetric measure; and $\Pr(k_a > k_b)$ for independent draws. All are computed on the log scale, as the density panels are drawn; the overlap is invariant to that choice because both densities acquire the same Jacobian.

5.6 Model and numerical validation

We validated the models against analytic limits and independent simulation. A deterministic one-day incubation period recovers the infection-anchored models after shifting the time index; as $k \rightarrow \infty$, SSE and SSI converge to Cori, while SSE-SO and SSI-SO converge to Cori-SO. Simulations from the explicit infection-and-incubation process agree with the onset-only representation of Proposition 1. Explicit reset simulations also agree with the risk formulas in Propositions 2 and 3, including the $R^* = 0$ limit in which $\text{RAT} = 0$ and RAC is the probability that the incubation pipeline is non-empty.

Particle filtering provides an independent check of the latent-state calculations, and particle marginal Metropolis–Hastings reproduces the parameter posteriors on synthetic data. Filtering and smoothed particle states are kept distinct because they correspond to different estimands (Section 4.5). Bridge-sampling estimates agree with deterministic quadrature for the low-dimensional models and with alternative estimators for the latent models. Finally, in the Poisson limit the analysis recovers the published reproduction-number posterior and Équateur risk curve of Thompson et al. (2024), after accounting for the one-day difference in time-indexing convention.

6 Results

Four analyses are reported. Analyses 1 and 2 compare the three infection-anchored models, with k held at the literature value 0.18 and then estimated; Analyses 3 and 4 add the onset-anchored models to SSE and SSI, again fixed and then estimated. Throughout, “the crossing” means the

first day on which a curve falls below a threshold *and stays there*. Values quoted at day 90 (4 July 2018) are read from the fitted curves partway down their descent, where they are furthest apart.

6.1 Mechanism, at a transplanted dispersion parameter

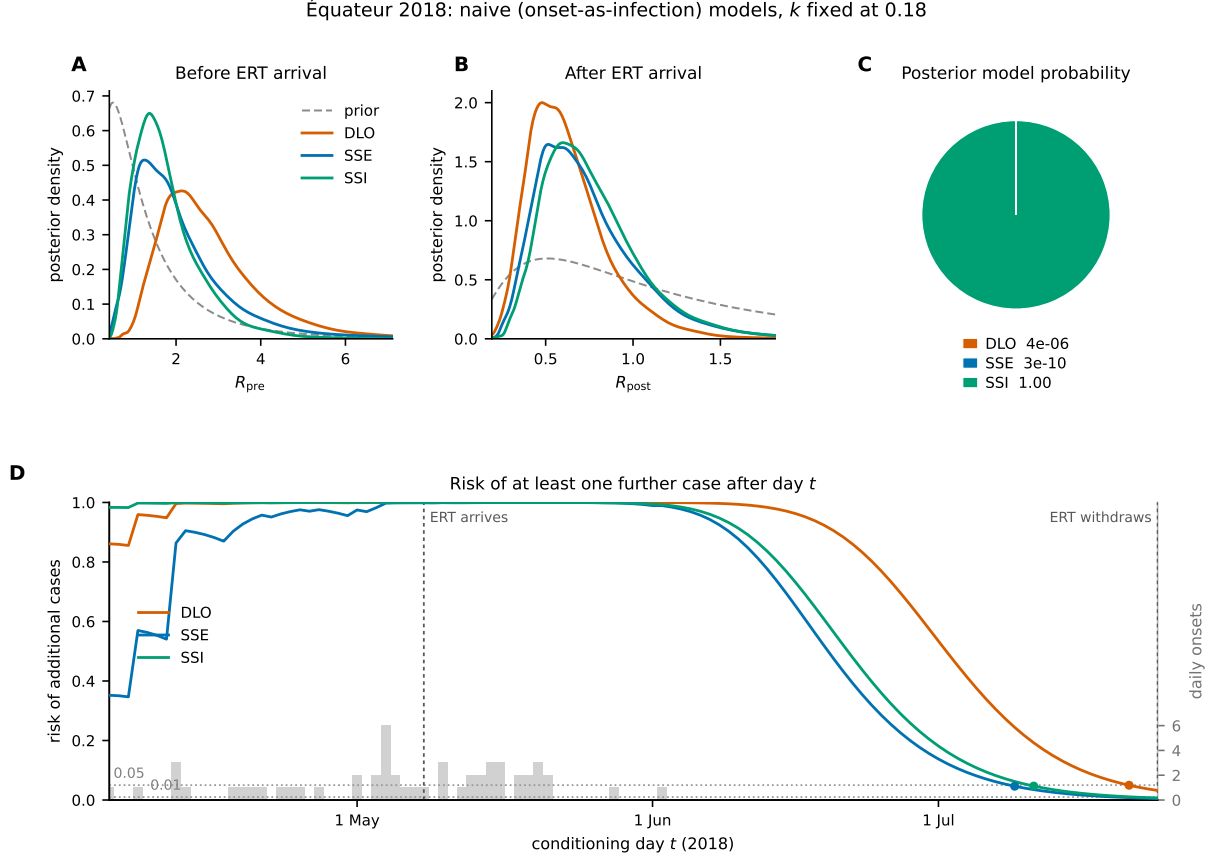


Figure 1: The mechanism of overdispersion changes the end-of-outbreak risk, at a dispersion parameter whose interpretation differs across models. DLO, SSE and SSI fitted to the Équateur onset series with k fixed at 0.18. **A**, **B**: posterior densities for R_{pre} and R_{post} , over the shared prior. **C**: posterior model probabilities under a uniform prior over the 3 models. Because the fixed value of k has a different interpretation in DLO, this panel compares the complete model specifications rather than mechanism alone. **D**: risk of additional cases against conditioning day, with daily onsets behind it, the ERT arrival and withdrawal marked, and the 0.05 and 0.01 decision thresholds drawn. The marker on each curve is its 0.05 crossing.

- **The same numerical k produces little overdispersion in DLO.** At day 90 the risk is 0.413 under DLO against 0.093 under SSE and 0.123 under SSI — a factor of more than four between models handed an identical dispersion parameter. This is the dilution bound (28) in practice: the future force of infection after the last case is thin and spread over the whole serial interval, which is exactly the regime in which a day-level k does almost nothing and DLO is pressed against its Poisson lower bound.
- **The consequence for a declaration is weeks, not days.** DLO's risk first settles below 0.05 on day 107 (21 July 2018), against day 95 for SSE and day 97 for SSI. At the 0.01 threshold DLO **never** settles within the analysis window, while SSE does so on day 106 and SSI on day 108. On the day the ERT actually withdrew, DLO still places 0.032 on a further case, against 0.005 and 0.007.
- **SSI lies on the same side of SSE as DLO does.** After the final case SSI's risk is *above* SSE's (0.123 against 0.093 at day 90), which reads backwards until one asks what each

model’s heterogeneity is attached to. SSI’s is a property of *people who have been observed*: 54 known individuals have now gone weeks without infecting anyone, which is direct evidence they were not the superspreading kind, and $\Lambda_Y(t)$ is revised down accordingly. SSE’s is a property of *events that have not happened yet*, about which a run of zeros says nothing, so SSE gets no such revision — its entire retained state is $\Lambda(t)$. At matched expected further cases, SSE’s residual risk is a rare burst and SSI’s a more likely singleton, and the question “any further case?” sees the second as the larger risk. The contrast is therefore not DLO against the other mechanisms: it is how many units the same $\Lambda(t)$ is divided over and whether each unit’s dispersion survives to time t . DLO divides over future days, SSI over past individuals, SSE not at all.

- **The ordering is a property of the question as much as of the models.** Asking instead whether transmission would re-establish reverses SSE and SSI on these same data, because a cluster seeds a new chain far more readily than a lone case. Both orderings are correct answers to different questions, and the end-of-outbreak criterion in use asks the first.
- **The model probabilities strongly favour SSI** — 1.00 against 4×10^{-6} for DLO and 3×10^{-10} for SSE, from log evidences of -84.8 , -97.1 and -106.8 nats. At a fixed k this partly reports that 0.18 is a better value for SSI than for the other two, which is the transplantation error restated rather than independent evidence about mechanism. Analysis 2 separates the two readings.
- **R_{post} is comparable across the three models** — 0.59 (0.30–1.20), 0.68 (0.34–1.48) and 0.72 (0.36–1.48) (posterior median and 95% credible interval) — so the divergence in the risk curves does not come from a different inferred effect of the intervention. The corresponding R_{pre} estimates are 2.55 (1.20–6.05), 1.81 (0.72–5.35) and 1.61 (0.73–3.87).
- **Monte-Carlo error does not account for the separation.** Over the case-free tail in which the crossings occur, the largest Monte-Carlo standard error on any of the three curves is 0.0015, two orders of magnitude below the differences above.

6.2 Mechanism, with the dispersion estimated

- **The main split in k is between individual- and day-level heterogeneity.** The posterior medians (95% credible intervals) are 0.378 (0.230–0.629) for DLO, 0.502 (0.307–0.853) for SSE and 0.141 (0.072–0.271) for SSI. DLO against SSI: overlap 0.094, median ratio 2.69, $\Pr(k_{\text{DLO}} > k_{\text{SSI}}) = 0.990$. SSE against SSI: overlap 0.032, ratio 3.57, probability 0.999. But DLO against SSE: overlap 0.589 and $\Pr(k_{\text{DLO}} > k_{\text{SSE}}) = 0.219$ — the two are barely distinguishable. **It is SSI that is the outlier, not DLO.**
- **The split is day-level against individual-level, which is where onset-as-infection bites hardest.** DLO and SSE both attach their excess variance to a *day*: DLO to the day’s aggregate incidence, SSE to the day’s pooled transmission through a freshly drawn λ_t . SSI attaches it to the *individual*. The series being fitted is onsets, and an incubation period of mean 11.4d and standard deviation 8.1d scatters each day’s infections forward over a wide kernel — so a model reading onsets as infections sees day-to-day variation that the convolution has already largely averaged out, and answers with a larger k (less overdispersion) than the infection process carried. SSI is close to invariant to this, because the convolution *regroups* individuals across days while the aggregate infectivity of n independent individuals is $\text{Gamma}(kn, k)$ whichever n they are: regrouping moves the cohorts, not the variance. Because all three models here are infection-anchored, this analysis alone cannot separate that explanation from the structural difference of Section 4.4; the onset-anchored comparison provides that test.
- **Only SSI’s posterior is compatible with the literature value.** SSI’s 95% credible interval, (0.072, 0.271), contains 0.18; DLO’s (0.230, 0.629) and SSE’s (0.307, 0.853) do not, and both medians sit above the prior’s own 97.5th percentile of 0.36, despite the informative

Équateur 2018: naïve (onset-as-infection) models, k estimated

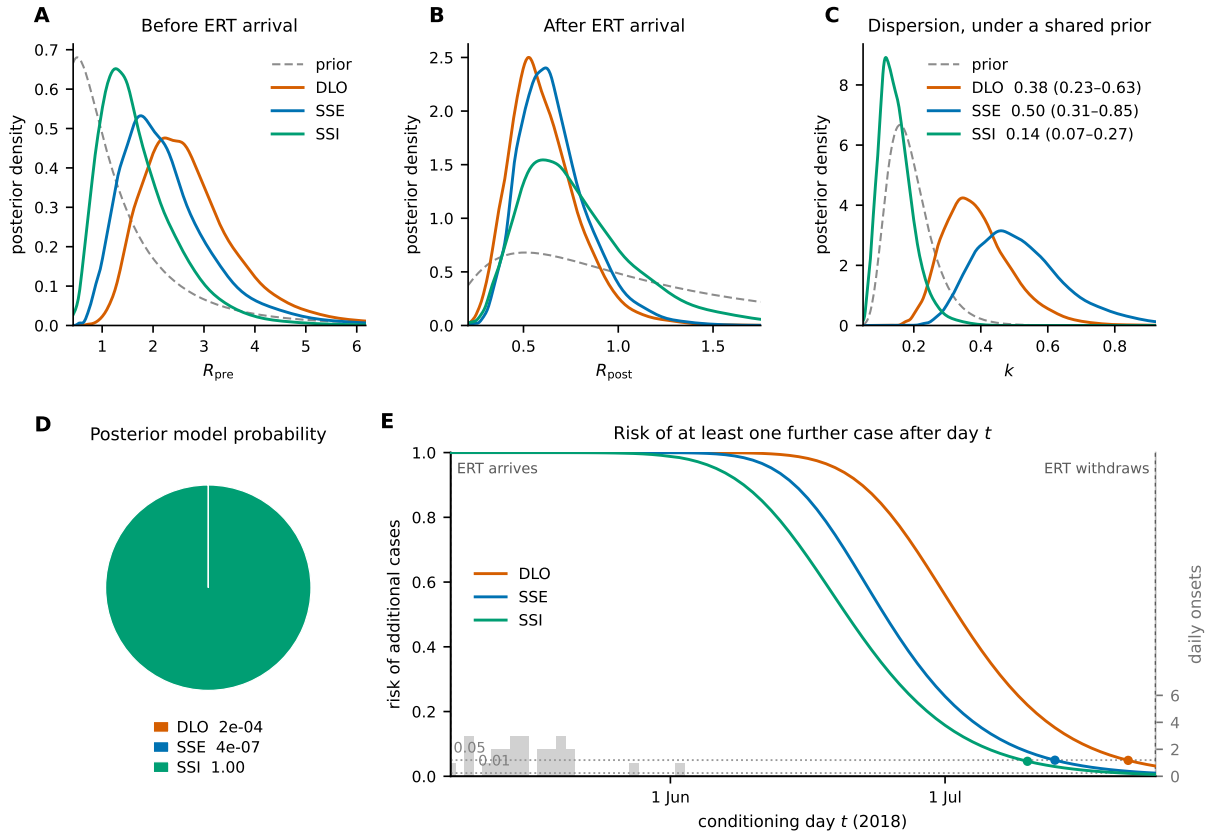


Figure 2: The models infer different dispersion parameters, and their risk curves remain separated. As Fig. 1, with k estimated. **C:** the three k posteriors over the prior they share, with each model's posterior median and 95% credible interval in the legend. **D:** posterior model probabilities. **E:** risk of additional cases, with the view starting at the ERT's arrival because all three curves sit flat at 1 until well past the last case.

prior.

- **The evidence says the same thing more sharply.** Letting k move is worth +7.41 nats to SSE and +3.90 to DLO, but only +0.26 to SSI. Analysis 1 was charging DLO and SSE for a value poorly supported under those models, whereas 0.18 is compatible with SSI on this series.
- **The risk gap persists when k is estimated.** At each model's inferred k , DLO still crosses 0.05 on day 107, against day 99 for SSE and day 96 for SSI, and still **never** reaches 0.01 inside the window, where SSE does on day 110 and SSI on day 107. Misapplying a literature k creates the gap; the day-level mechanism sustains it.
- **The posterior model probabilities keep their direction** — SSI 1.00, DLO 2×10^{-4} , SSE 4×10^{-7} . Because each model has its own inferred dispersion, this comparison is not confounded by imposing the same numerical value of k .

6.3 Onset anchoring

Équateur 2018: onset anchoring compared, k fixed at 0.18

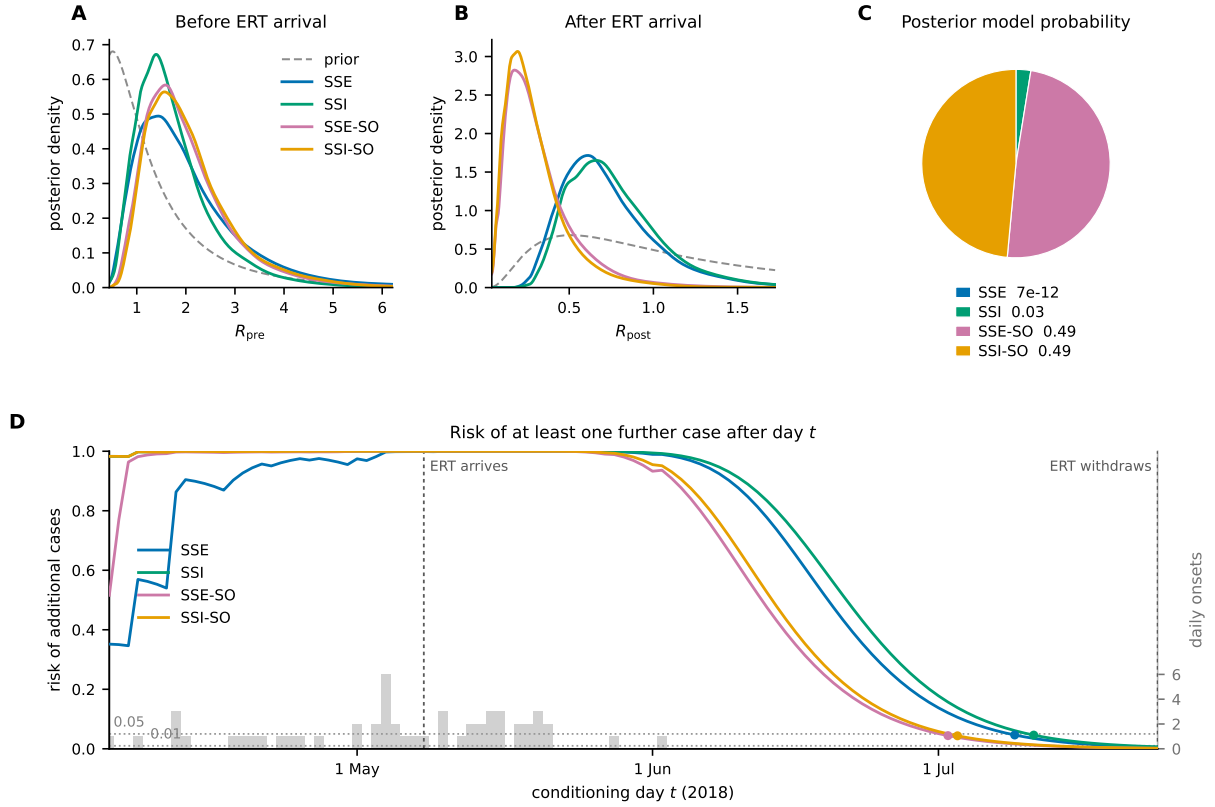


Figure 3: Onset anchoring at a fixed dispersion parameter. SSE, SSI, SSE-SO and SSI-SO with k held at 0.18; panels as in Fig. 1, with a uniform prior over 4 models in C. All four models are driven by the same onset-to-onset serial interval, so the comparison isolates anchoring.

- **Onset anchoring advances both crossings by 8–12 days.** With k estimated (Fig. 4), SSE-SO crosses 0.05 on day 87 against SSE's day 99, an advance of 12 days, and 0.01 on day 98 against day 110, an advance of 12 days. SSI-SO advances SSI's crossings by 8 and 8 days respectively (day 88 against day 96, and day 99 against day 107). At a fixed k (Fig. 3) the same comparisons give 7 and 7 days for SSE-SO against SSE, and 8 and 9 days for SSI-SO against SSI. The displacement is on the scale of the mean incubation period, 11.4 d.

Équateur 2018: onset anchoring compared, k estimated

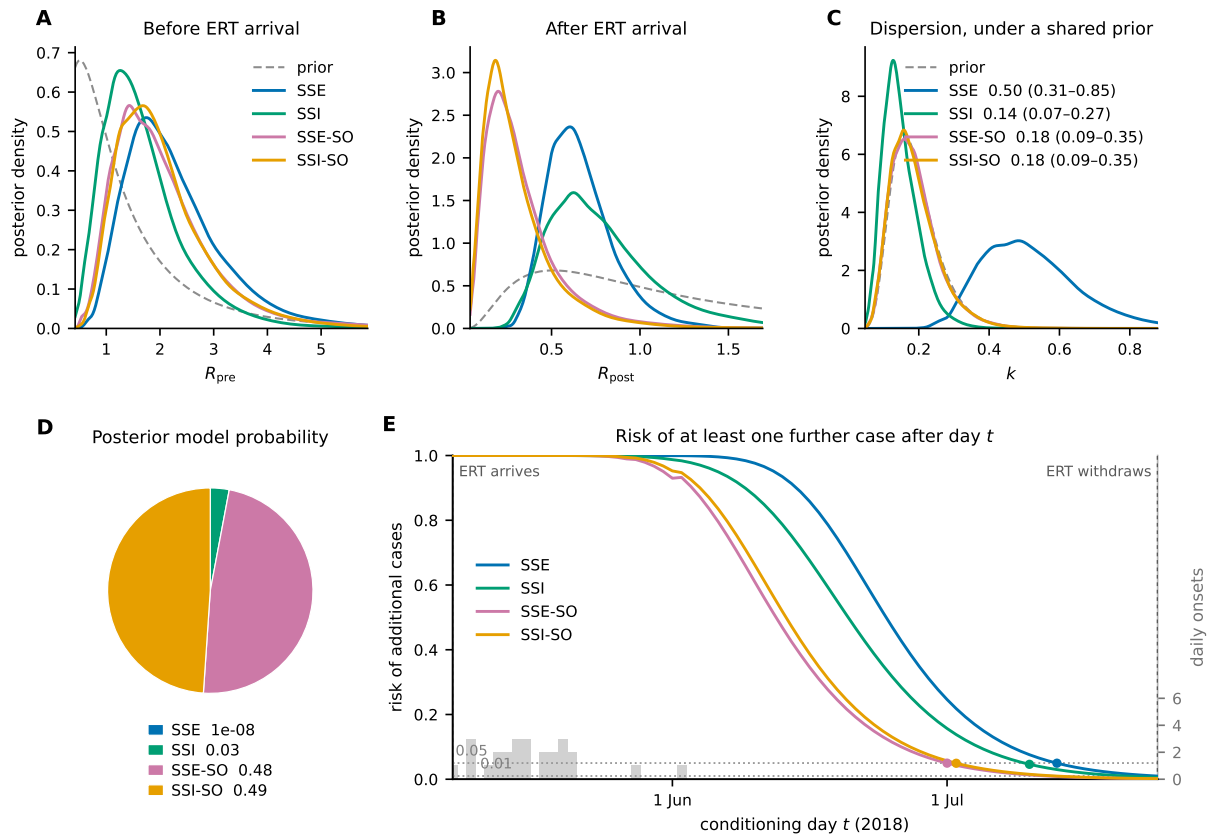


Figure 4: Onset anchoring with the dispersion estimated. Panels as in Fig. 2. In panel C, SSE-SO’s k posterior shifts below SSE’s and towards the literature estimate, whereas SSI-SO remains close to SSI. This contrast supports attenuation of day-level, but not individual-level, heterogeneity when onsets are treated as infections.

- **Why the direction is *earlier*, and why it is not a shorter generation time.** All four models share the same onset-to-onset serial interval; what changes is which side of the conditioning day a transmission falls on once that interval is cut into transmission and incubation pieces. Proposition 3 makes this explicit: the onset-anchored models place infection at transmission time, so infections already generated under the historical R_{post} enter $M(t)$ and stay in the incubation pipeline when control is reset, while the remaining TOST opportunity $W(t)$ of older onset cohorts is already largely spent. The infection-anchored models have no pipeline, move the infection decision to the later symptom-onset date, and apply the reset R^* to it, which makes the residual state look younger and drain later.
- **The infection-anchored convention more than doubles the inferred R_{post} .** With k estimated the posterior medians (95% credible intervals) are 0.63 (0.37–1.11) for SSE and 0.73 (0.35–1.60) for SSI, against 0.28 (0.08–0.95) for SSE-SO and 0.26 (0.08–0.89) for SSI-SO. The infection-anchored convention applies the day-33 switch to *onsets*, so post-arrival onsets that arose from pre-arrival infections are charged to the controlled period. R_{pre} is far more stable across the four — 2.09 (0.98–4.77), 1.54 (0.65–3.67), 1.86 (0.84–4.61) and 1.87 (0.87–4.41) — consistent with an effect on *when* the switch lands rather than on the epidemic’s early growth.
- **The results support attenuation of day-level dispersion.** SSE-SO’s k posterior, 0.178 (0.088–0.351), sits below infection-anchored SSE’s 0.501 (0.306–0.847) and towards the literature value: median ratio 2.81, overlap 0.089, $\Pr(k_{\text{SSE}} > k_{\text{SSE-SO}}) = 0.992$. SSI moves far less — 0.140 (0.072–0.271) against SSI-SO’s 0.175 (0.088–0.353), overlap 0.750. Incubation convolution attenuates *day-level* dispersion when onsets are read as infections; individual-level dispersion is comparatively stable under regrouping.
- **The two onset-anchored mechanisms infer similar k values.** SSE-SO and SSI-SO agree almost exactly — overlap 0.980, median ratio 1.02, $\Pr(k_{\text{SSE-SO}} > k_{\text{SSI-SO}}) = 0.505$ — where their infection-anchored counterparts differ by a factor of 3.58 at an overlap of 0.034. The event-versus-individual gap of Section 6.2 is therefore largely attributable to temporal conflation in this outbreak.
- **The onset-anchored models are strongly preferred, and are indistinguishable from each other.** With k estimated the posterior model probabilities are 0.48 for SSE-SO and 0.49 for SSI-SO, against 0.03 for SSI and 1×10^{-8} for SSE; the log evidences are -81.8 , -81.8 , -84.6 and -99.4 nats. The fixed- k analysis gives the same ordering (0.49, 0.49, 0.03, 7×10^{-12}).
- **Estimating k adds little evidence for the onset-anchored models.** Releasing k is worth $+0.01$ nats to SSE-SO and $+0.04$ to SSI-SO — within their own estimation error — against $+7.41$ for infection-anchored SSE. Their k posteriors already sit on 0.18, so fixing it there costs them nothing.
- **The 8–12-day advance is specific to this analysis.** Its direction is conditional on the reset and switch conventions, the outbreak history and the posterior; it is not a general result for arbitrary time series or interventions.

6.4 Cases against transmission: the incubation pipeline

- **The pipeline is worth about a quarter of a unit of probability at its widest.** With k estimated, RAC exceeds RAT by at most 0.269 for SSE-SO, on day 64 (8 June 2018), and by at most 0.240 for SSI-SO, on day 65. The gap peaks shortly after the last observed onset (day 58), when the pipeline is largest relative to the transmission still to come, and closes as it drains.
- **It delays a case-based declaration by most of a week.** With k estimated, SSE-SO’s RAT settles below 0.05 on day 81 and below 0.01 on day 92, against day 87 and day 98 for RAC — later by 6 and 6 days. For SSI-SO the same comparison gives 5 and 6 days (RAT

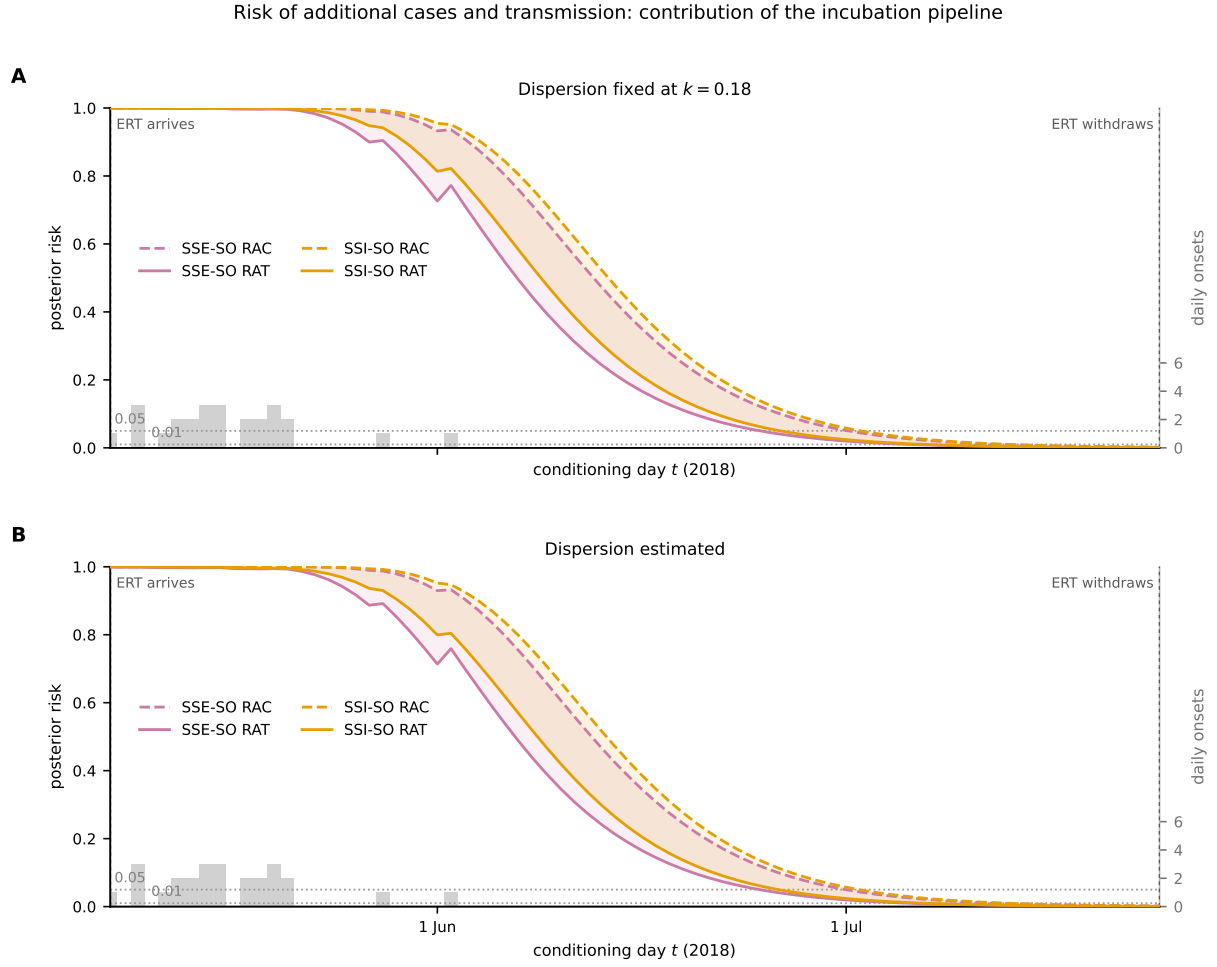


Figure 5: The gap between the risk of additional cases and the risk of additional transmission is the incubation pipeline, quantified. SSE-SO and SSI-SO with k fixed at 0.18 (A) and estimated (B). Dashed: RAC. Solid: RAT. The shaded region is generated by the incubation-pipeline term $M(t)$ in Proposition 3: individuals already infected on the conditioning day but not yet symptomatic. Under the three infection-anchored models these two curves coincide by assumption.

Table 1: Threshold crossings of the risk of additional cases across the four analyses: the first day on which the curve settles below the threshold, with the calendar date. The ERT withdrew on day 110 (24 July 2018); “never” means the curve had not settled below the threshold by then.

Analysis	Model	RAC < 0.05		RAC < 0.01	
		day	date	day	date
1: infection-anchored, $k = 0.18$	DLO	107	21 July 2018	never	never
	SSE	95	9 July 2018	106	20 July 2018
	SSI	97	11 July 2018	108	22 July 2018
2: infection-anchored, k estimated	DLO	107	21 July 2018	never	never
	SSE	99	13 July 2018	110	24 July 2018
	SSI	96	10 July 2018	107	21 July 2018
3: onset, $k = 0.18$	SSE	95	9 July 2018	106	20 July 2018
	SSI	97	11 July 2018	108	22 July 2018
	SSE-SO	88	2 July 2018	99	13 July 2018
	SSI-SO	89	3 July 2018	99	13 July 2018
4: onset, k estimated	SSE	99	13 July 2018	110	24 July 2018
	SSI	96	10 July 2018	107	21 July 2018
	SSE-SO	87	1 July 2018	98	12 July 2018
	SSI-SO	88	2 July 2018	99	13 July 2018

on days 83 and 93 against RAC on days 88 and 99).

- **The two quantities answer different questions, and the difference is decision-relevant.** RAT isolates what relaxing interventions does; RAC is what a surveillance system would actually see. A criterion written on cases keeps interventions in place while the pipeline drains, even though transmission risk has already settled — and an infection-anchored model, which identifies the two by assumption, cannot represent that distinction at all.
- **Fixing k changes very little here**, as the near-identical panels show: the widest gap is 0.269 for SSE-SO and 0.236 for SSI-SO, against 0.269 and 0.240 when it is estimated — a consequence of the onset-anchored k posteriors already sitting on the literature value.

7 Planned sensitivity analyses

The following analyses remain to be completed before drawing conclusions about robustness.

1. **Outbreak-specific serial interval** (mean 19.46 d, standard deviation 6.08 d) — *infection-anchored models only*. Its lighter tail may reduce the estimated risk after a long case-free period. It is incompatible with the onset-anchored models under the independent incubation/TOST decomposition because its variance is smaller than the incubation variance (Section 2.2).
2. **Incubation/TOST decomposition.** Re-estimate SSE-SO and SSI-SO under alternative published Ebola incubation distributions while holding the onset-to-onset serial interval fixed. This will quantify dependence on a decomposition that the serial interval alone does not identify.
3. **Filtering rather than smoothed latent states.** Integrate the filtering comparison of Section 4.5 over parameter uncertainty for all three latent-variable models.
4. **Shifted R -switch for the infection-anchored models.** The onset-anchoring comparison confounds model structure with the indexing convention for the intervention. Moving the switch in the infection-anchored models back by $\mathbb{E}[A] = 11.4$ days will help separate the effect of intervention indexing from the effect of model structure.

5. **Non-empty initial incubation pipeline** for the onset-anchored models. Seed one or two already-infected individuals at day 0 with residual incubation drawn from f^{inc} , and measure the effect on R_{pre} and the risk curve. This assesses sensitivity to whether the first observed case is the epidemiological index case (Section 2.1).
6. **Prior sensitivity**, particularly for the informative prior on k , whose interpretation differs between models.

A The two forms of the onset-anchored models are equivalent

This appendix proves Proposition 1 and gives the additional argument needed for (13). Throughout, the explicit infection-process form is as defined in Section 3.3: infections J_t are drawn from the model’s offspring distribution, and the J_u infections of day u are allocated to onset days by the incubation split (9), of which only the totals D_t are observed.

Proof of Proposition 1

By hypothesis $J_t \mid E_t \sim \text{Poisson}(E_t)$, with E_t measurable after the day- t onset cohort and any associated latent state have been determined.

Step 1 — Poisson thinning. Multinomial splitting of a Poisson count yields *independent* Poisson parts: conditional on E_u , $N_{u,a} \sim \text{Poisson}(E_u f_a^{\text{inc}})$ independently over $a \geq 1$, with $J_u = \sum_a N_{u,a}$ recovered as their sum. The latent J_u therefore carries no information beyond the parts and can be dropped. (This is also the step that makes the incubation pipeline of (23) Poisson.)

Step 2 — Poisson superposition. $D_t = \sum_{a \geq 1} N_{t-a,a}$ is a sum of independent Poissons, hence $D_t \mid (E_u)_{u < t} \sim \text{Poisson}(\sum_{a \geq 1} f_a^{\text{inc}} E_{t-a})$, which is (10).

Step 3 — the joint law, not just the marginal. Because f^{inc} has no mass at lag 0, E_{t-a} for $a \geq 1$ depends only on onset cohorts and associated states dated no later than $t-1$, so the display above is precisely the conditional law $p(D_t \mid D_{0:t-1})$. The parts $\{N_{u,a} : u+a=t\}$ composing D_t are disjoint from those composing any other $D_{t'}$, so distinct days’ onsets are conditionally independent given the E s. Chaining $p(D_{0:T}) = \prod_t p(D_t \mid D_{0:t-1})$ gives equality of the full joint laws. $\square$

Why the hypothesis covers all three models

Cori-SO and SSI-SO satisfy it directly, with $E_t = R_t \lambda_t$ and $E_t = R_t \lambda_t^Y$ respectively; under SSI-SO the Gamma infectivities are explicit in both forms and the thinning and superposition arguments do not touch them.

SSE-SO needs one further step, because its offspring law (13) is negative binomial rather than Poisson. If $\tilde{\lambda}_t \mid \lambda_t \sim \text{Gamma}(k\lambda_t, k)$ and $J_t \mid \tilde{\lambda}_t \sim \text{Poisson}(R_t \tilde{\lambda}_t)$, then $R_t \tilde{\lambda}_t \sim \text{Gamma}(k\lambda_t, k/R_t)$ and, marginally, $J_t \sim \text{NB}(\text{mean} = R_t \lambda_t, \text{disp} = k\lambda_t)$ — which is exactly (13). So (14) merely *un-marginalises* the negative binomial into an explicit per-day Gamma random effect, after which J_t is Poisson conditional on $\tilde{\lambda}_t$ and the proposition applies with $E_t = R_t \tilde{\lambda}_t$. The $\tilde{\lambda}_t$ are independent across t given (λ_t) , and each λ_t is a function of $D_{\leq t}$, so the day- t conditional still depends only on $D_{0:t-1}$ once $\tilde{\lambda}_{<t}$ is integrated out. $\square$

B Data and code availability

The accompanying repository contains the daily onset series, analysis configuration, source code, posterior samples, derived quantities, figures and this manuscript. The full analysis is reproducible from the onset data through a documented Snakemake workflow. Numerical values in the manuscript are generated from the saved analysis outputs so that text and figures remain synchronised. The repository also includes the simulation, particle-method and marginal-likelihood validation analyses described in Section 5.6.

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